EC2

Key Points & Mini Project

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Amazon Web Services (AWS) Elastic Compute Cloud (EC2) is one of the fundamental services for anyone looking to delve into the world of cloud computing. EC2 provides scalable and flexible virtual computing resources in the AWS cloud. In this article, we will explore key points about EC2, especially those relevant to certification exams, and then embark on a hands-on mini project to create a web application using EC2 instances.

EC2 — Elastic Compute Cloud

Key Points for AWS EC2

Understanding EC2 Instances

- EC2 instances are virtual servers in the cloud.
- They come in various types optimized for different workloads (e.g., compute, memory, GPU).
- Knowing how to choose the right instance type for your application is crucial.

EC2 Instance Lifecycle

- Instances have different states: pending, running, stopping, stopped, terminated.
- You are billed for instances in the “running” state.

Security Groups and Network ACLs

- Security Groups act as firewalls for EC2 instances.
- Network Access Control Lists (NACLs) control traffic at the subnet level.
- Both are essential for controlling inbound and outbound traffic.

EC2 Storage Options

- Instances can have both root (boot) and additional EBS volumes.
- Understanding the differences between instance store and EBS storage is vital.
- Snapshotting EBS volumes for backup is a common practice.

Elastic IP Addresses

- Elastic IPs are static IP addresses that can be associated with EC2 instances.
- Useful for applications that require a fixed public IP.

User Data and Metadata

- User data can be used to run scripts during instance launch.
- Metadata provides information about the instance (e.g., instance type, public IP).

Auto Scaling and Load Balancing

- Auto Scaling ensures that you have the right number of instances based on traffic.
- Elastic Load Balancer (ELB) distributes traffic across multiple instances for high availability.

EC2 Instance Storage Optimization

- Utilize instance store volumes for temporary data.
- For data persistence, attach EBS volumes.

Instance Types and Use Cases

- Understand the types (e.g., T2, M5, P3) and their best use cases (e.g., general-purpose, memory-optimized, GPU instances).

IAM Roles for EC2

- Assign IAM roles to instances to grant them access to AWS services without using access keys.

Mini Project

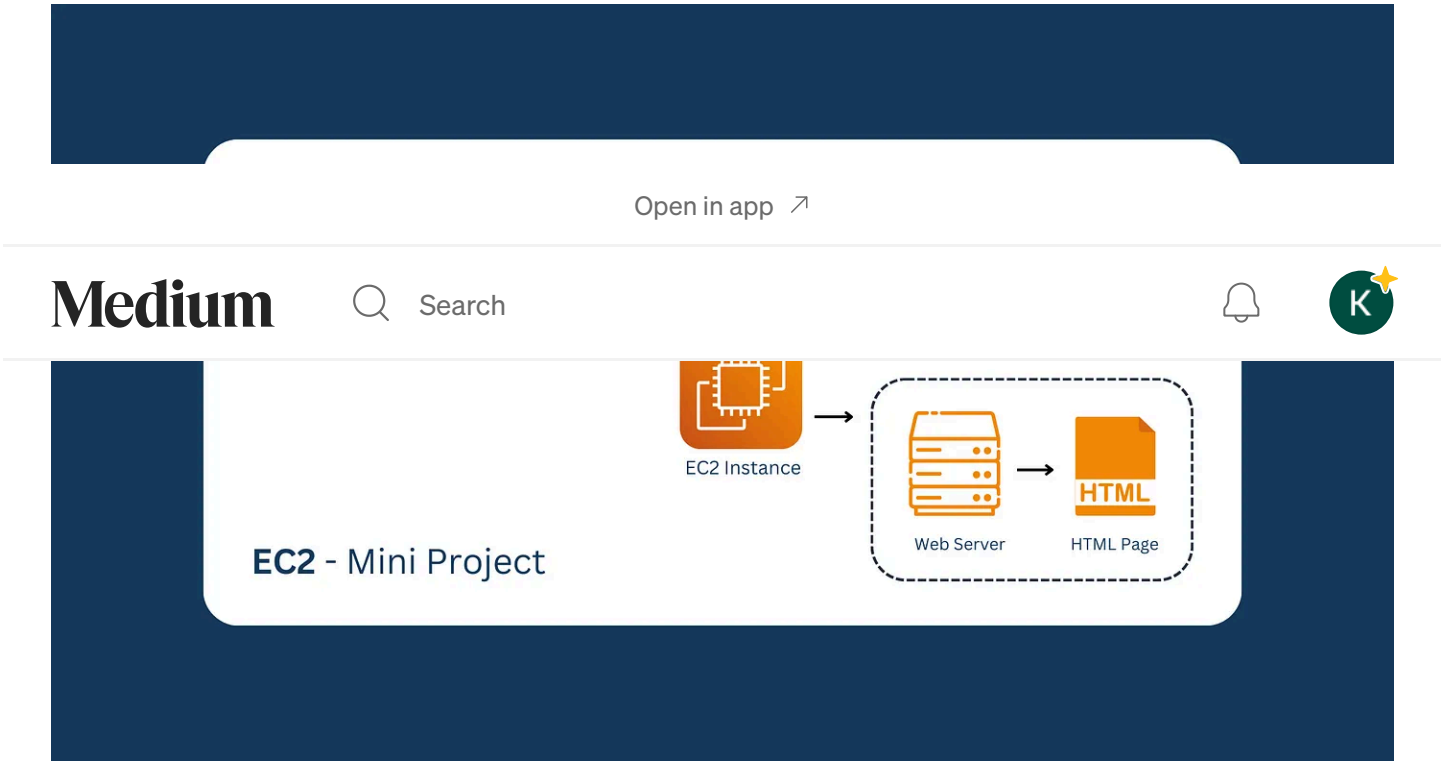
> *Creating a Web Application using EC2 Instances*

Now, let's put our knowledge into practice with a mini project. We will create a simple web application using EC2 instances:

Project Objective

Build a basic web application that displays "Happy Learning, Mr.CloudExplorer!" when accessed via a web browser.

Architecture Diagram



EC2 Web Application-Architecture Diagram

Steps

1. **Launch an EC2 Instance**

- Sign in to your AWS Management Console.
- Navigate to the EC2 Dashboard.
- Launch a single EC2 instance.
- Choose an appropriate instance type and configure security groups to allow HTTP traffic.

2. **Set Up Web Server**

- Install a web server (e.g., Apache, Nginx) on the EC2 instance.
- Create a simple HTML file with the “Happy Learning, Mr.CloudExplorer!” message.

3. **Assign a Public IP**

- If the instance does not have a public IP assigned by default, allocate an Elastic IP address and associate it with the instance. This gives your instance a static public IP.

4. **Test the Application**

- Access the instance’s public IP or DNS name in a web browser to see the “Happy Learning, Mr.CloudExplorer!” message.

5. **Cleanup**

- Terminate the EC2 instance to avoid incurring additional charges.

AWS-Projects / EC2 - Elastic Compute Cloud

Hands on Steps to re-Create.

github.com

By following these simplified steps, you'll gain a basic understanding of how to set up a web application on a single EC2 instance. This project is an excellent starting point for beginners to AWS and EC2 and can be a foundation for more complex architectures as you gain more experience.

Happy Learning ;)

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